

# Digital Communications

## OFDM

### Orthogonal Frequency Division Multiplexing

#### 1st part



# Outline

- **Wireless Channel**
- – Fading
  - Fast Fading
  - Frequency Selectivity
- **OFDM – Orthogonal Frequency Division Multiplexing**
  - Multichannel modulation with orthogonal pulses
  - Transmitting with many carriers
  - Frequency domain description of the channel



# Introduction

- Modulation:
  - A mapping of the information on changes in the carrier phase, amplitude or frequency or a combination of both
- Multiplexing:
  - A method of sharing a bandwidth with other independent data channels
- OFDM is a combination of modulation & multiplexing



# Introduction

- Multiplexing generally refers to independent channels, those produced by independent sources
- So, it is a question of how to share the spectrum with these users
- In OFDM the question of multiplexing is applied to independent signals, but these are a sub-set of the one main signal
- In OFDM the signal itself is first split into independent signals, modulated by data and then re-multiplexed to create the OFDM carrier



# Introduction

- OFDM is a special case of Frequency Division Multiplex (FDM)
- As an analogy, a FDM channel is like a water flow out of a faucet, in contrast the OFDM signal is like a shower



- In a faucet all water comes in one big stream and cannot be sub-divided
- OFDM shower is made up of a lot of little streams

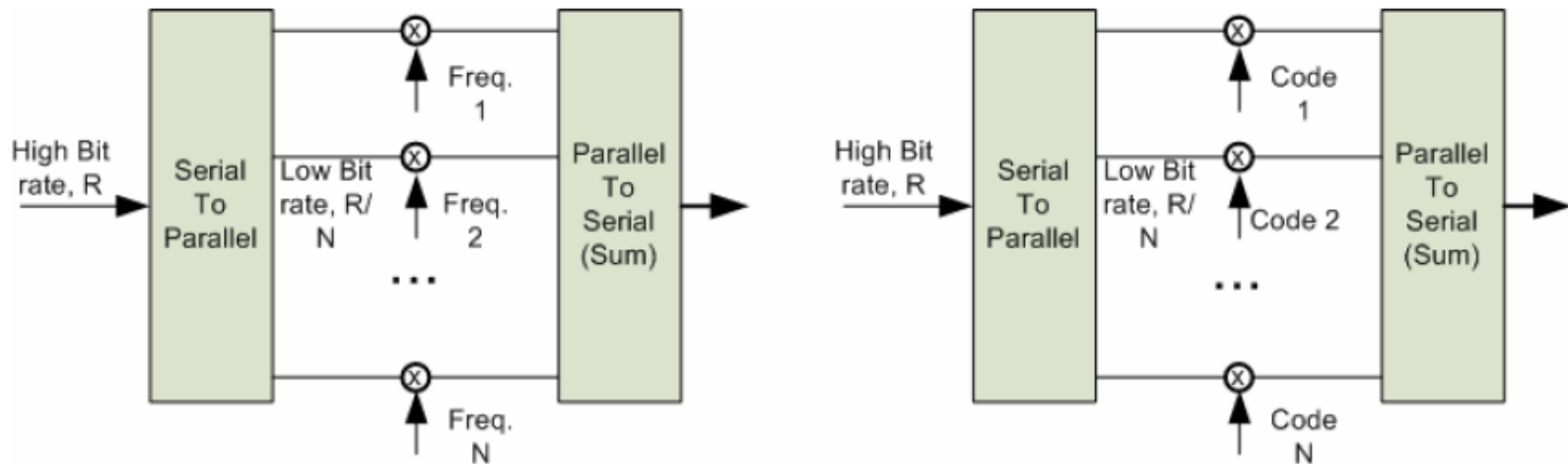
# Introduction

- Think about what the advantage might be of one over the other?
- One obvious one is that if I put my thumb over the faucet hole, I can stop the water flow, but I cannot do the same for the shower
- So, although both do the same thing, they respond differently to interference



# Introduction

- The independent sub-channels can be multiplexed by Frequency Division Multiplexing (FDM), called multi-carrier transmission or it can be based on a Code Division Multiplex (CDM), in this case it is called multi-code transmission



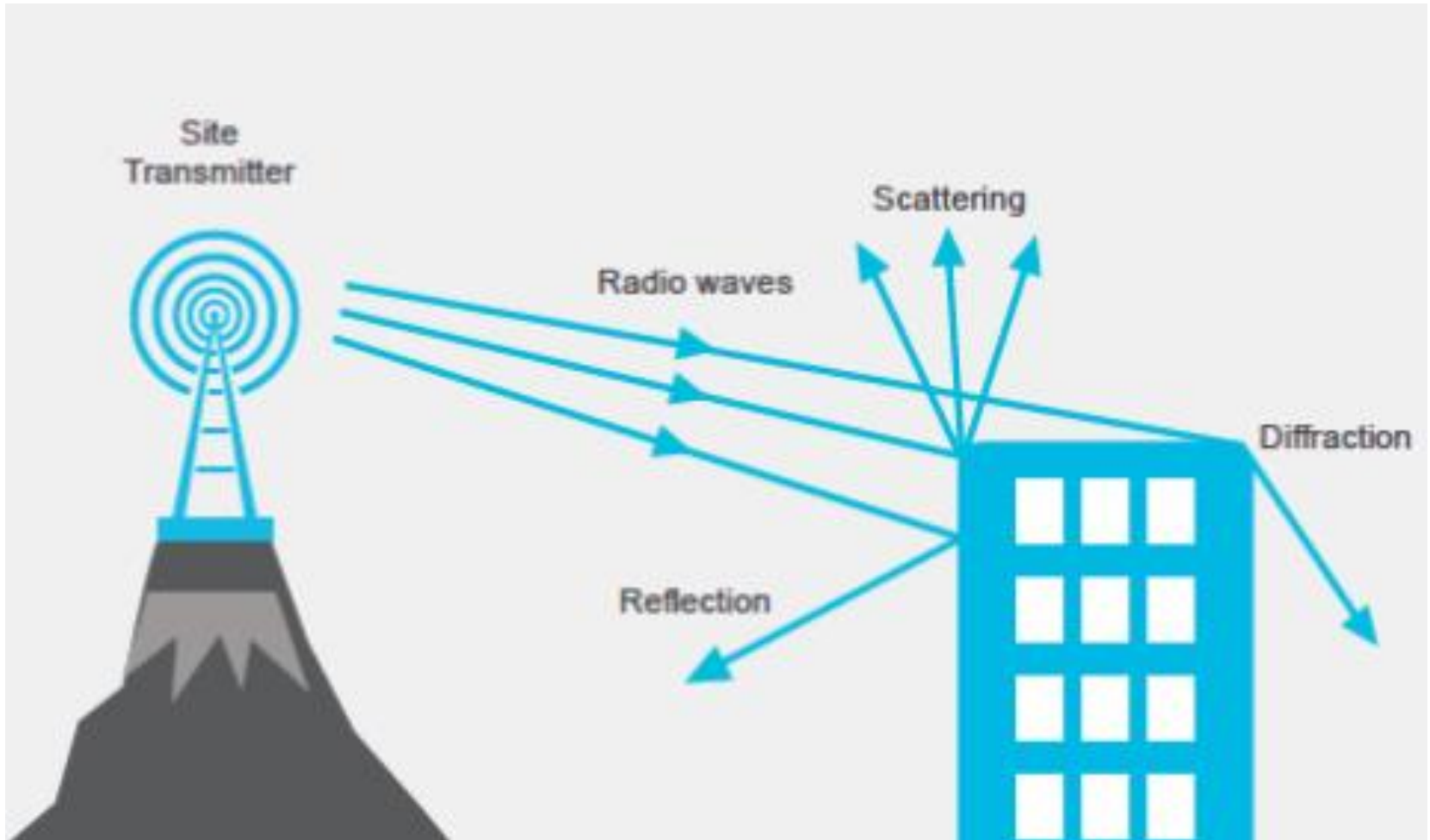
# Wireless Channels

- In wireless channels the transmitted signal can travel different distances before arrive to the receiver
- This is known as the multipath effect and can cause:
  - Fading
  - Inter-Symbol-Interference





# Propagation



# Propagation

- Diffraction
  - Occurs when the wave is obstructed by an obstacle and secondary waves are generated
- Scattering
  - When the wavelength is about the same size of the blocking objects. The incident wave is reflected in many directions
- Reflection
  - If the wavelength is smaller than the dimensions of the reflecting object

# Sinusoids and Phasors

- Suppose that the transmitted signal  $s(t)$  is a sinusoid with frequency  $f_0$
- In a static situation, the received signal  $r(t)$  can be modelled as a sum of sinusoids with different amplitudes and phases:

$$r(t) = \sum_{n=1}^N a_n s(t - \tau_n) = \sum_{n=1}^N a_n \cos(2\pi f_0(t - \tau_n))$$

$$r(t) = \sum_{n=1}^N a_n \cos(2\pi f_0 t + \phi_n)$$



# Sinusoids and Phasors

- Using the complex notation, we can write the previous equation in the form:

$$r(t) = \sum_{n=1}^N a_n \Re\{e^{j(\omega_0 t + \phi_n)}\} = \Re\{e^{j\omega_0 t} A e^{j\phi}\}$$

$$r(t) = A \cos(\omega_0 t + \phi)$$

- where:

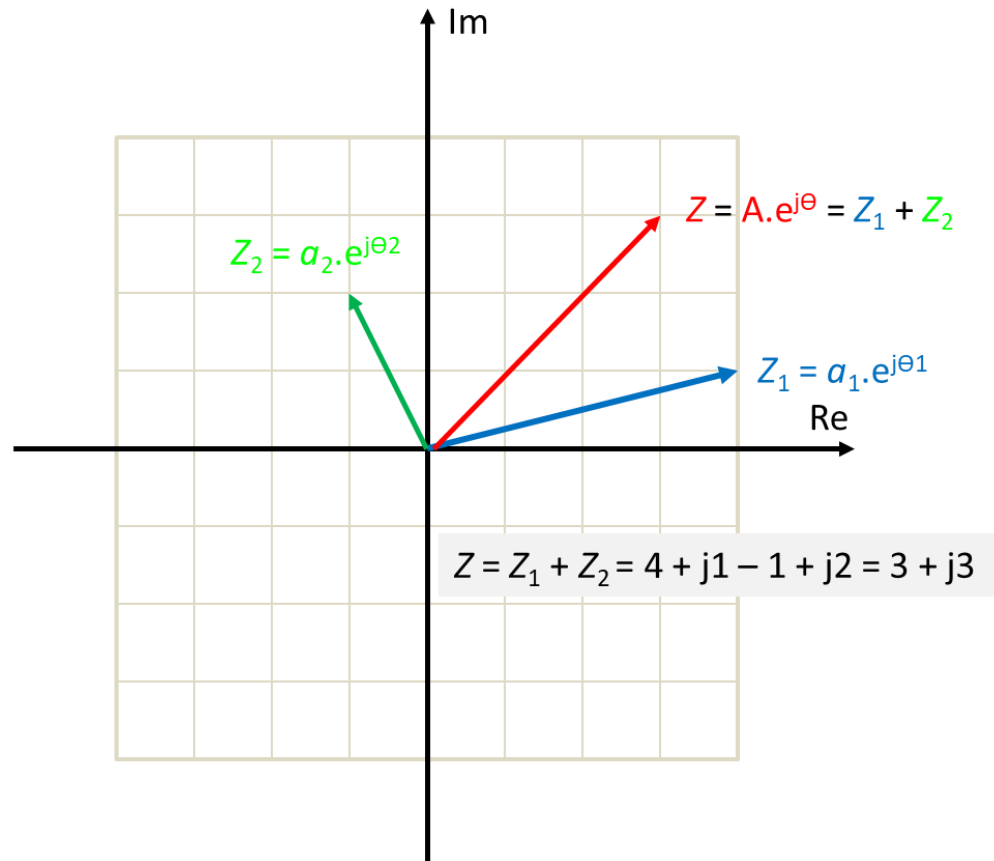
$$A e^{j\phi} = \sum_{n=1}^N a_n e^{j\phi_n}$$

- The sum of  $N$  sinusoids, with the same frequency but random amplitude and phase, is another sinusoid



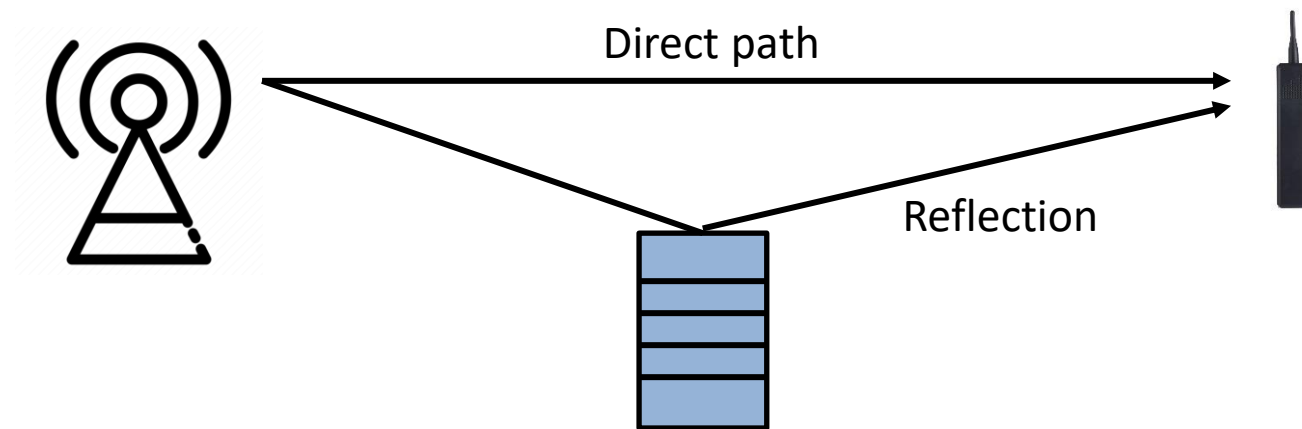
# Summing Sinusoids

- Geometric Interpretation with Phasors



# Summing Sinusoids

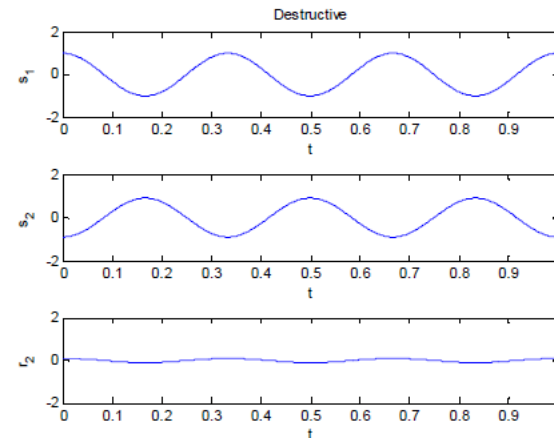
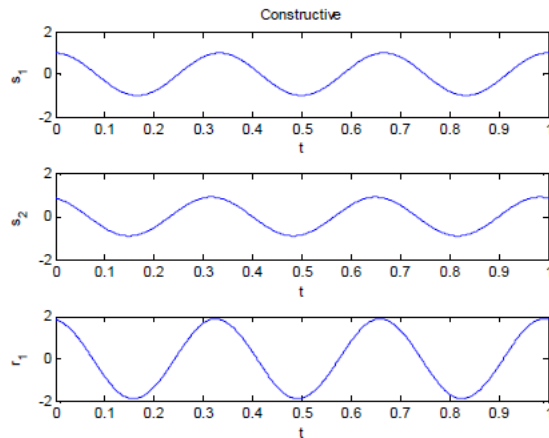
- Simple example:
  - Consider just a single reflection
  - The received signal would be just the direct path and one reflection



$$r(t) = a_1 s(t - \tau_1) + a_2 s(t - \tau_2)$$

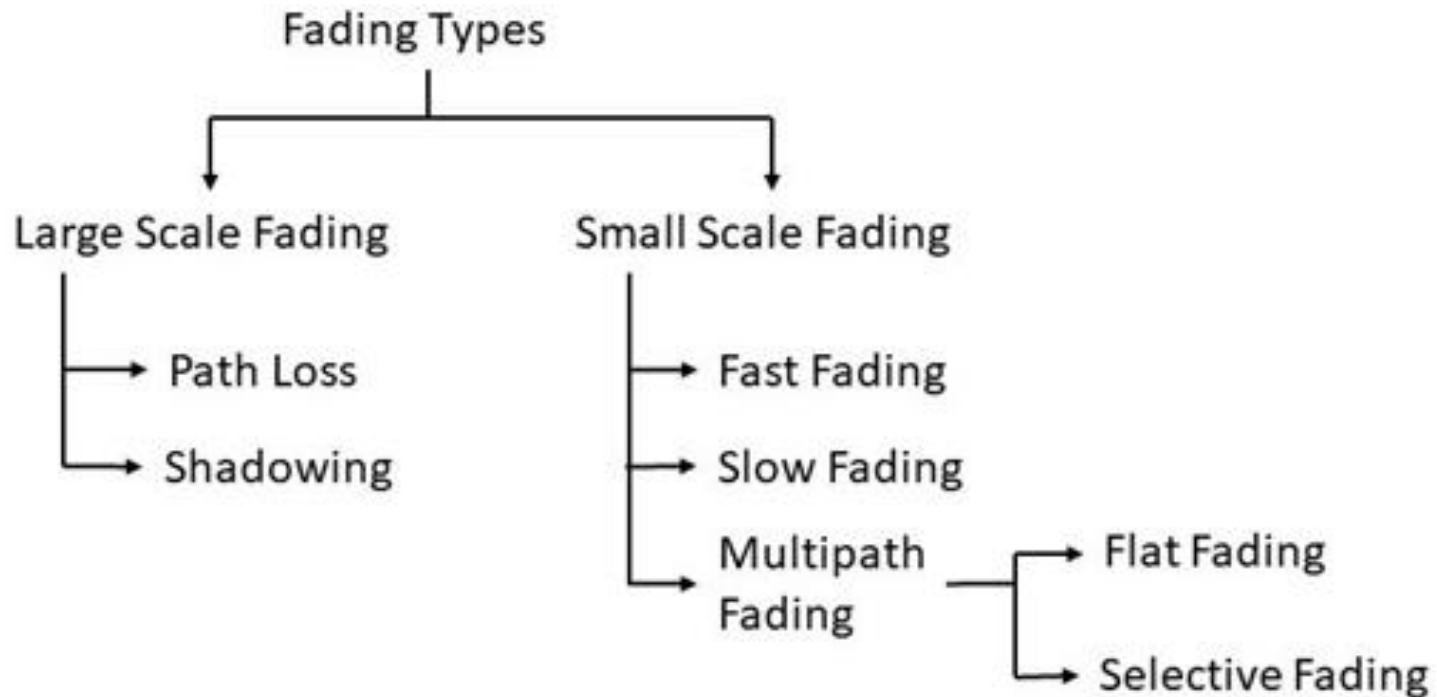
# Summing Sinusoids

- Fading:
  - Due to the different paths, the two received signals present different phase and amplitude
  - As we can see their sum can be destructive
  - With more reflections the amplitude of the received signal can deeply change for different receiver positions



# Fading

- Fading is a phenomenon that occurs due to varying parameters and conditions of the channel during wireless propagation





# Large Scale Fading

- This refers to the attenuation of signal power due to obstacles between the transmitter and receiver
  - It also covers the attenuation and fluctuations of signal when the signal is transmitted over a long distance
- Path Loss:
  - It refers to the attenuation when a signal is transmitted over large distances
  - Wireless signals spread as they propagate through the medium and as the distance increases, the energy per unit area starts decreasing and it is independent of the type of transmitter and medium



# Large Scale Fading

- Shadowing:
  - This refers to the loss in signal power due to the obstructions in the path of propagation
    - There are a few ways in which shadowing effects can minimize signal loss. One that is most effective, is to have a Line-Of-Sight propagation
    - Shadowing losses also depend on the frequency of the EM wave
    - EM Waves can penetrate through various surfaces but at the cost of loss in power i.e. signal attenuation
    - The losses depend on the type of the surface and frequency of the signal. Generally, the penetration power of a signal is directly proportional to the frequency of the signal



# Small Scale Fading

- This refers to the fluctuations in signal strength and phase over short distance and small duration of time
  - It is also called Rayleigh Fading
- Small Scale Fading affects almost all forms of wireless communication and overcoming them is a necessity to increase efficiency and decrease error

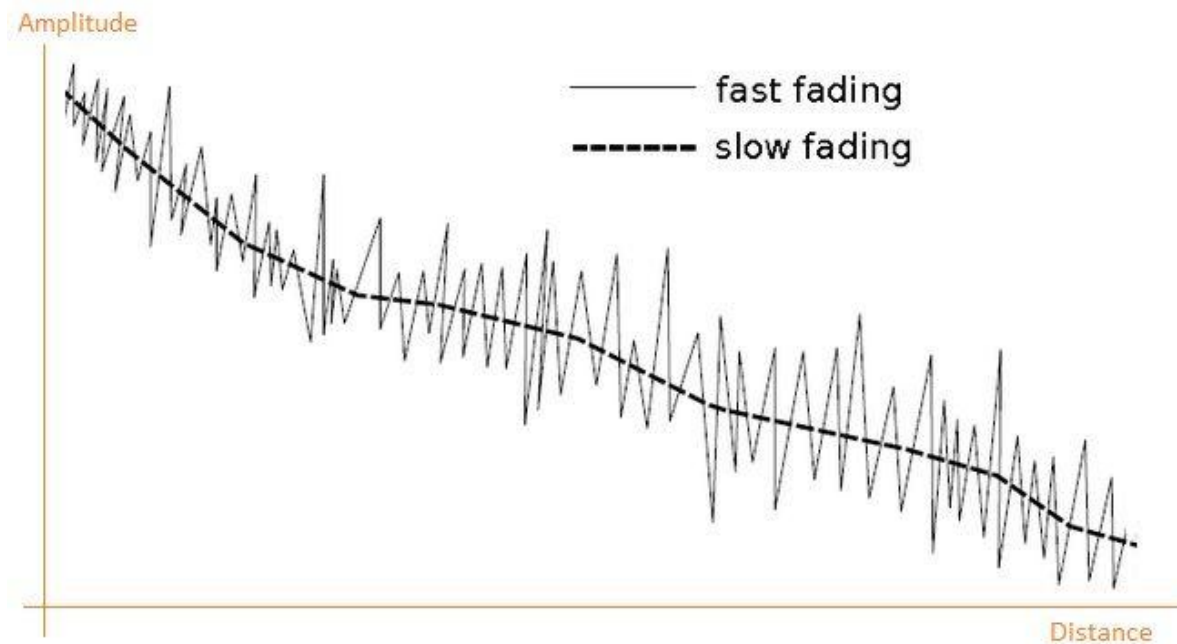


# Small Scale Fading

- Fast Fading:
  - It occurs mainly due to reflections for surfaces and movement of transmitter or receiver
    - High Doppler spread is observed in the fast fading with Doppler bandwidth comparable to or greater than the bandwidth of the signal and the channel variations are as fast or faster than the signal variations
    - It causes linear distortions in the shape of the baseband signal and creates Inter Symbol Interference
    - One way to remove ISI is adaptive equalization

# Small Scale Fading

- Fast fading originates due to effects of constructive and destructive interference patterns which is caused due to multipath



# Small Scale Fading

- Slow Fading:
  - It occurs mainly due to shadowing where large buildings or geographical structures obstruct the LOS
    - Low doppler spread is observed in Slow Fading with the doppler bandwidth being smaller compared to the bandwidth of the signal and the channel variations are slow relative to the signal variations
    - It results in reduction of SNR which can be overcome using error correction and receiver diversity techniques



# Small Scale Fading

- Multipath Fading:
  - It occurs when a signal reaches the receiver from various paths i.e. when multipath propagation takes place
  - Multipath fading can affect all ranges of frequencies starting from low frequency to microwave and beyond
  - It affects both the amplitude and the phase of the signal causing phase distortions and ISI



# Small Scale Fading

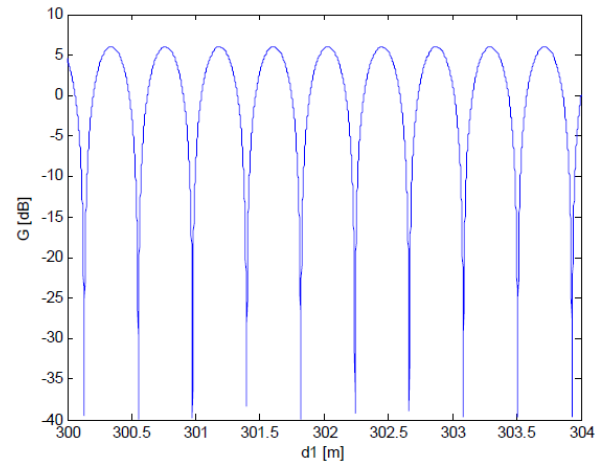
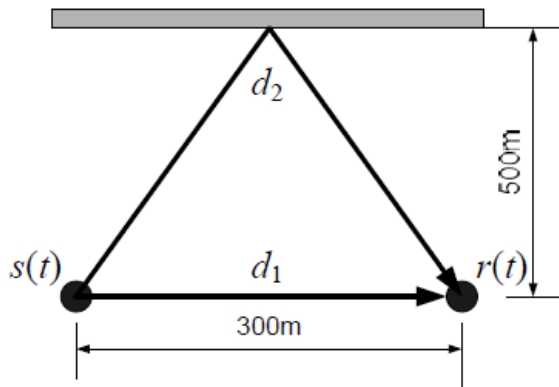
- Multipath fading can affect signal transmission in two ways:
  - Flat Fading:
    - All frequency components get affected almost equally
      - Causes the amplitude to fluctuate over a period of time
  - Selective Fading:
    - Refers to multipath fading when the selected frequency component of the signal is affected
      - Selected frequencies will have increased error and attenuation as compared to other frequency components of the same signal





# Fast Fading

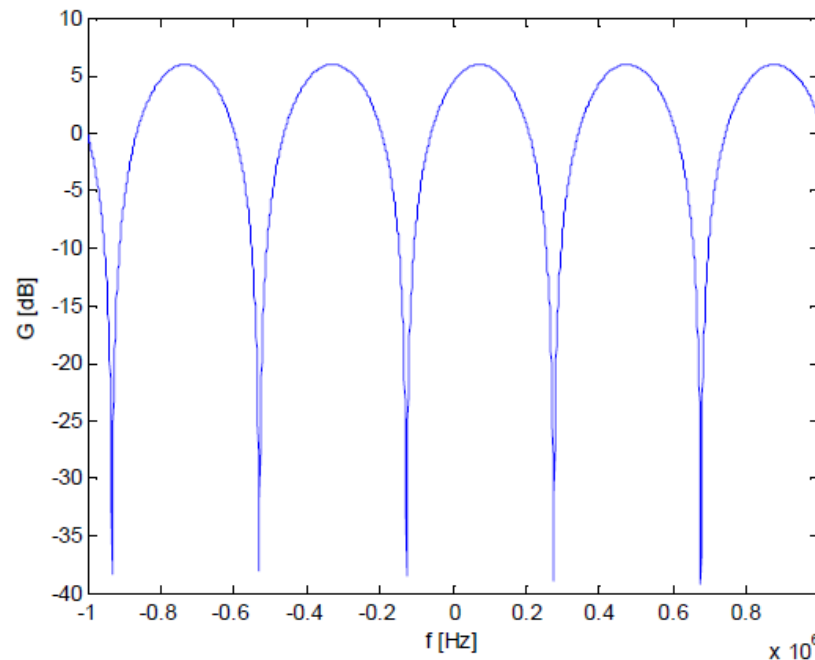
- In this example we consider a carrier with a frequency of 1GHz and an obstacle at 500m



- Small changes in the distance between the emitter and the receiver can deeply change the amplitude of the received signal

# Frequency Selectivity

- For the same example, the envelope of the received signal also changes with the frequency of the signal



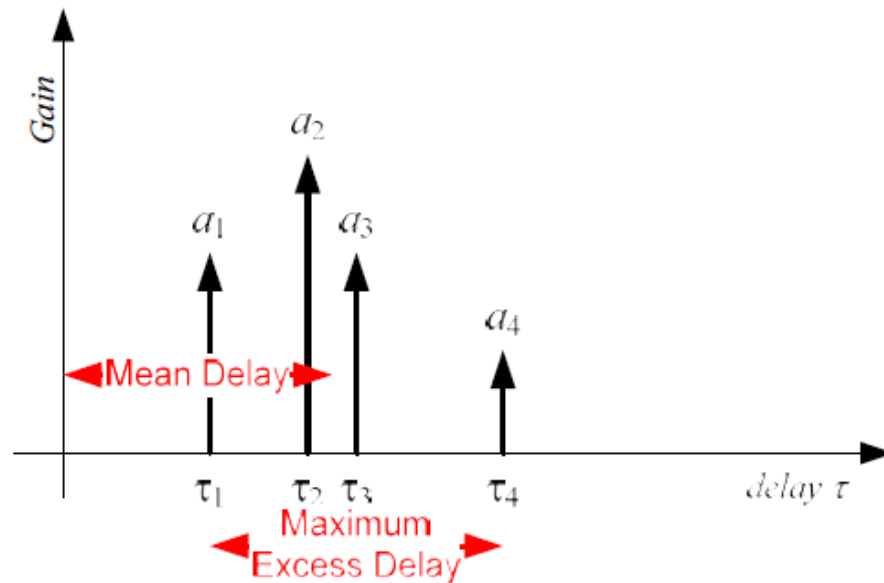
# Frequency Selectivity

- If the bandwidth of the signal is too narrow when compared with the width of the nulls, the received signal can be deeply attenuated
- For wireless communications, new modulation techniques were developed to overcome these problems



# Multipath Propagation

- Time dispersive channel



## Maximum Excess Delay

The relative delay of the last multipath component to the first arriving component

## Mean Delay

$$\bar{\tau} = \frac{\sum_{n=1}^N a_n^2 \tau_n}{\sum_{n=1}^N a_n^2}$$

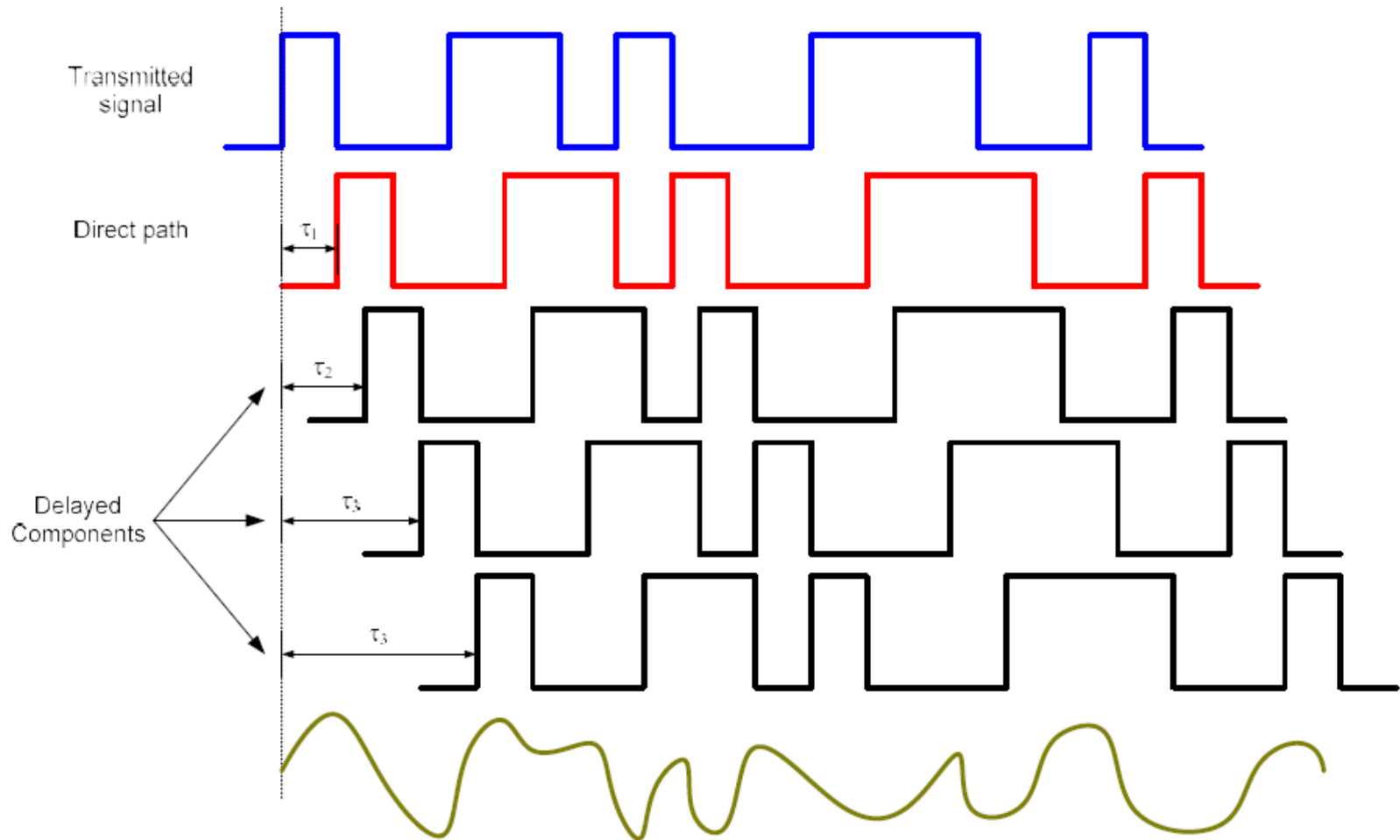


# Multipath Propagation

- Time dispersive channel
  - The impulse response can be modelled by a sum of delayed impulses with different amplitudes
  - For high data rates inter-symbol interference appears
  - The impulse response can change in time due to changes in the position of the mobile receiver
  - Classical approach: adaptive equalization of the channel

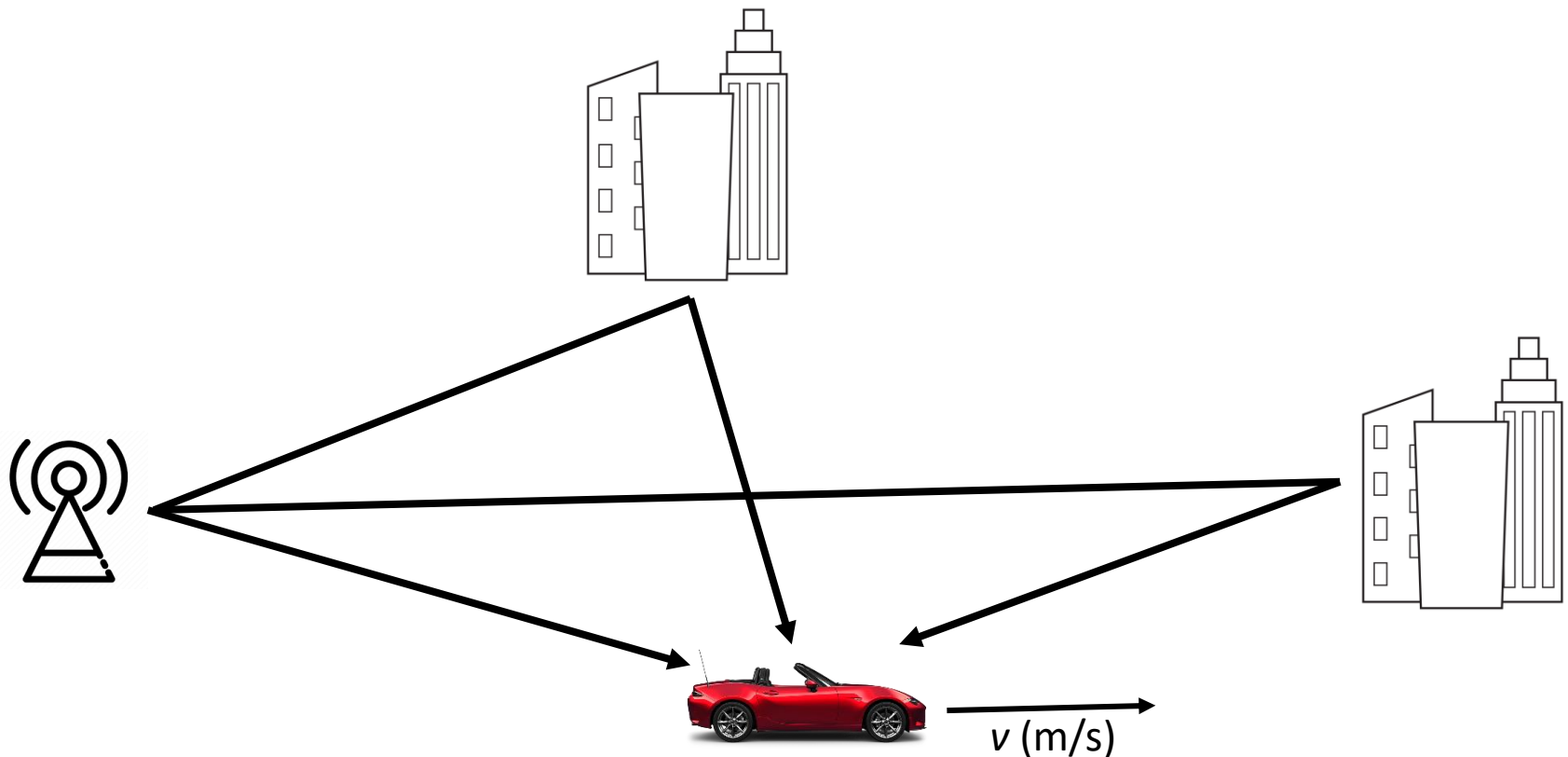


# Inter-Symbol-Interference

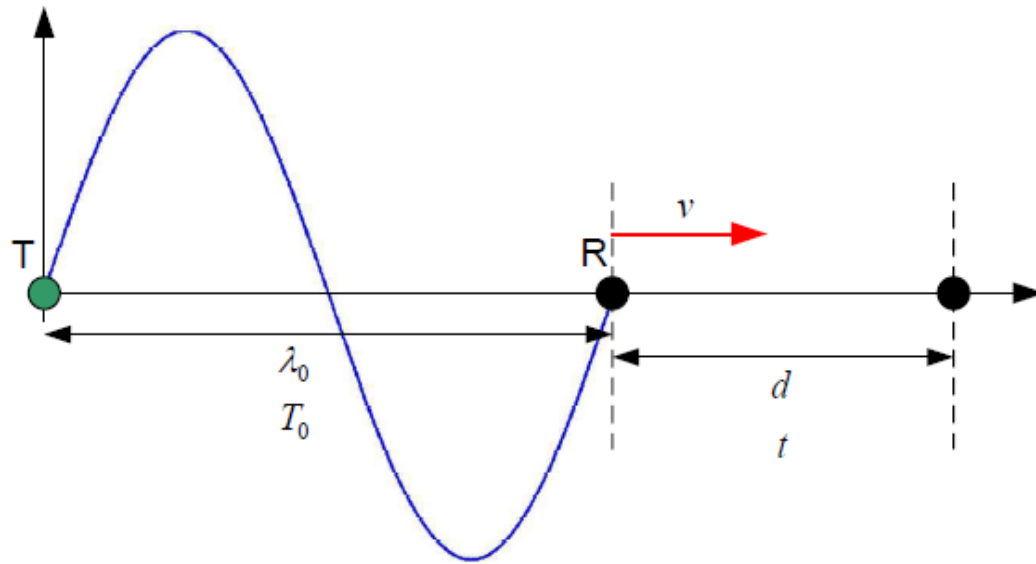


# Doppler Frequency Shift

- Let's consider the following scenario:



# Doppler Frequency Shift



T – Transmitter

R – Receiver

$T_0$  – Wave period

$\lambda_0$  – Wavelength

$c$  – Speed of light

$v$  – Receiver Speed

$$c = \frac{\lambda_0}{T_0} = \lambda_0 f_0$$

$$v = \frac{d}{t}$$

$$\lambda = \lambda_0 + d = \lambda_0 + vt$$

During a period  $T_0$  we have

$$\lambda = \lambda_0 + vT_0 = \lambda_0 + v \frac{\lambda_0}{c}$$

$$\lambda = \lambda_0 \left( 1 + \frac{v}{c} \right)$$





# Doppler Frequency Shift

- As the speed of propagation  $c$  does not change, we can write the previous equation as:

$$\frac{c}{f} = \frac{c}{f_0} \left( 1 + \frac{v}{c} \right)$$

- Finally, we have:

$$f = f_0 \left( \frac{c}{c + v} \right)$$

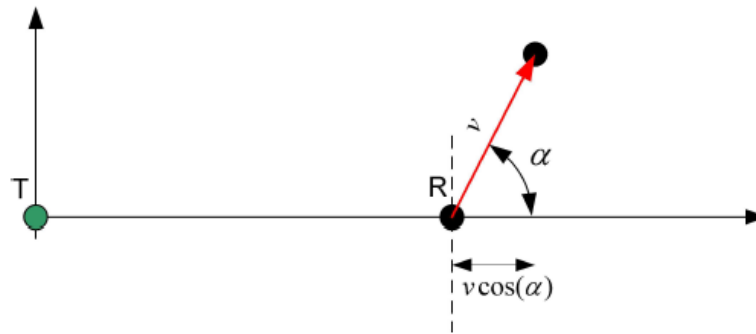
- The Doppler Shift  $f_D$  can be defined as the difference:

$$f_D = f_0 - f = f_0 - f_0 \left( \frac{c}{c + v} \right) = f_0 \left( \frac{v}{v + c} \right) \approx f_0 \frac{v}{c} = \frac{v}{\lambda_0}$$



# Doppler Frequency Shift

- If there is an angle  $\alpha$  between the speed direction of the receiver



- If  $\alpha > \pm 90$  the Doppler Shift will be negative
- For a vehicle traveling at a speed  $v$ , we can evaluate the worst case for the Doppler Shift:

$$f_D = f_0 \pm f = \pm f_0 \frac{v}{c}$$

# Doppler Frequency Shift

- Example:
- Consider a car traveling at a speed of 120km/h and a carrier frequency of 900MHz. Then we will have a Doppler Shift of:

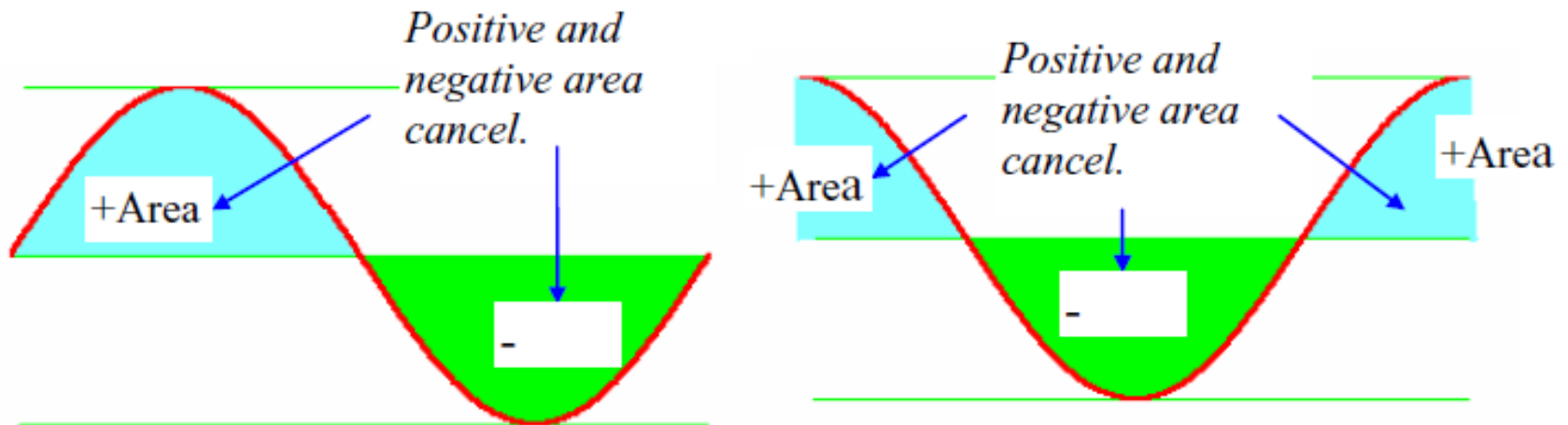
$$f_D = \pm f_0 \frac{v}{c} = \pm 900 \times 10^6 \frac{120}{1080 \times 10^6} = \pm 100 \text{ Hz}$$

– Where the speed of light is:  $c = 3 \cdot 10^8 \text{ m/s} = 1080 \times 10^6 \text{ km/h}$



# OFDM

- The importance of being orthogonal
  - The main concept in OFDM is orthogonality of the sub-carriers. Since the carriers are all sine/cosine wave, we know that area under one period of a sine or a cosine wave is zero. This is easily shown:



# OFDM

- The importance of being orthogonal
  - Let's take a sine wave of frequency  $m$  and multiply it by a sinusoid (sine or a cosine) of a frequency  $n$ , where both  $m$  and  $n$  are integers
  - The integral or the area under this product is given by:

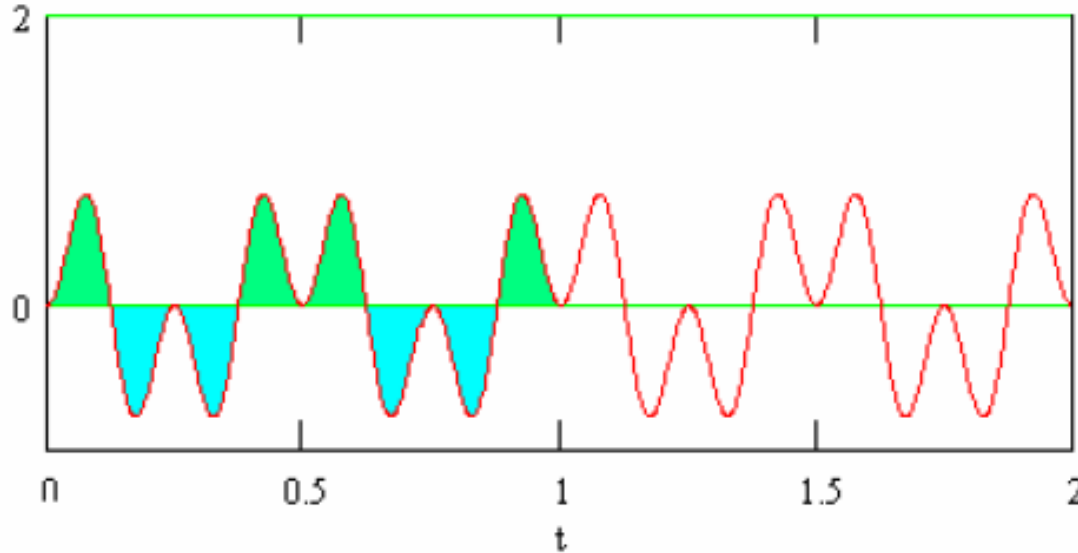
$$f(t) = \sin m\omega t \times \sin n\omega t = \frac{1}{2}\cos(m-n)\omega t - \frac{1}{2}\cos(m+n)\omega t$$

- These two components are each a sinusoid, so the integral is equal to zero over one period:

$$I = \int_0^{2\pi} \frac{1}{2}\cos(m-n)\omega t - \int_0^{2\pi} \frac{1}{2}\cos(m+n)\omega t = 0 - 0$$

# OFDM

- The importance of being orthogonal
  - We conclude that when we multiply a sinusoid of frequency  $n$  by a sinusoid of frequency  $m/n$ , the area under the product is zero



# OFDM

- The importance of being orthogonal
  - In general, for all integers  $n$  and  $m$ ,  $\sin(mx)$ ,  $\cos(mx)$ ,  $\sin(nx)$ ,  $\cos(nx)$ , are all orthogonal to each other
  - These frequencies are called harmonics
  - This idea is key to understanding OFDM
  - The orthogonality allows simultaneous transmission on a lot of sub-carriers in a tight frequency space without interference from each other
    - In essence this is similar to CDMA, where codes are used to make data sequences independent (also orthogonal) which allows many independent users to transmit in same space successfully



# FDM

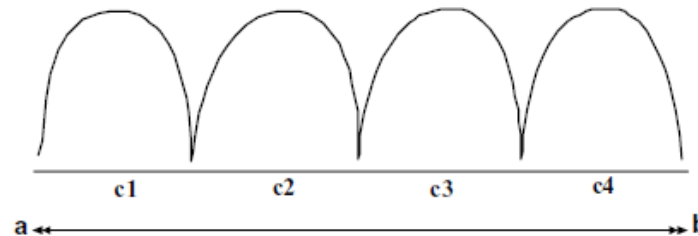
- OFDM is a special case of FDM
  - just as it says in its name: OFDM
- Let's first look at what a Frequency Division Multiplexing (FDM) is
- If we have a bandwidth that goes from frequency  $a$  to  $b$ , we can subdivide this into a frequency space of four equal spaces





# FDM

- In frequency space the modulated carriers would look like this:



- The frequencies  $a$  and  $b$  can be anything, integer or non-integer
  - since no relationship is implied between  $a$  and  $b$
  - Same is true of the carrier center frequencies
    - which are based on frequencies that do not have any special relationship to each other

# FDM

- But, what if frequency  $c_1$  and  $c_n$  were such that for any  $n$ , an integer, the following holds:
  - $c_n = n \times c_1$
- So that:
  - $c_2 = 2 \times c_1$
  - $c_3 = 3 \times c_1$
  - $c_4 = 4 \times c_1$
- All three of these frequencies are harmonic to  $c_1$



# FDM

- Since these carriers are orthogonal to each other, when added together, they do not interfere with each other
- In FDM, since we do not generally have frequencies that follow the above relationship, we get interference from neighbor carriers
- To provide adjacent channel interference protection, signals are moved further apart

# FDM

- The symbols rate that can be carried by a PSK carrier of bandwidth  $B$ , is given by:

$$R_s = 2B_l = B_p$$

- Where  $B_l$  is lowpass bandwidth and  $B_p$ , the passband bandwidth
- This relationship assumes a perfect Nyquist filtering with rolloff = 0.0



# FDM

- Since this is unachievable, we use root raised cosine filtering which for a roll-off of  $\alpha$  gives the following relationship:

$$R_s = \frac{B_p}{1 + \alpha}$$

- If we need three carriers, each of data rate = 20 Mbps, and with  $R_s = 20$  and  $B = 20 \times 1.25 = 25$  MHz
- Each carrier may be placed  $(25 + 2.5)$  27.5 MHz apart allowing for a 10% guard band
- The frequencies would not be orthogonal but in FDM we don't care about this. It's the guard band that helps keep interference under control



# OFDM

- In OFDM we have  $N$  carriers
- Let's examine the following bit sequence we wish to transmit and show the development of the OFDM signal using 4 sub-carriers:
  - First few bits are: 1, 1, -1, -1, 1, 1, 1, -1, 1, -1, -1, -1, -1, 1, -1, -1, -1, -1, 1,...
- The signal has a symbol rate of 1 and sampling frequency is 1 sample per symbol, so each transition is a bit



# OFDM

- Now write these bits in rows of fours, since this demonstration will use only four sub-carriers
- We have effectively done a serial to parallel conversion:

| c1 | c2 | c3 | c4 |
|----|----|----|----|
| 1  | 1  | -1 | -1 |
| 1  | 1  | 1  | -1 |
| 1  | -1 | -1 | -1 |
| -1 | 1  | -1 | -1 |
| -1 | 1  | 1  | -1 |
| -1 | -1 | 1  | 1  |



# OFDM

- Each column represents the bits that will be carried by one sub-carrier
- Let's start with the first carrier,  $c_1$ . What should be its frequency?
- From the Nyquist sampling theorem, we know that smallest frequency that can convey information has to be twice the information rate
- In this case, the information rate per carrier will be 1/4 or 1 symbol per second total for all 4 carriers





# OFDM

- So the smallest frequency that can carry a bit rate of  $1/4$  is  $1/2$  Hz
- Choosing  $1/2$  Hz as my starting frequency, then my harmonics are: 1,  $3/2$  and 2 Hz
  - But we choose 1 Hz for convenience
  - If we choose  $7/8$  Hz to start, the harmonics are:  $7/4$ ,  $21/8$ ,  $7/2$  Hz
- Choosing BPSK as our modulation scheme
  - For QPSK, we have to double the bit rate while keeping the symbol rate the same

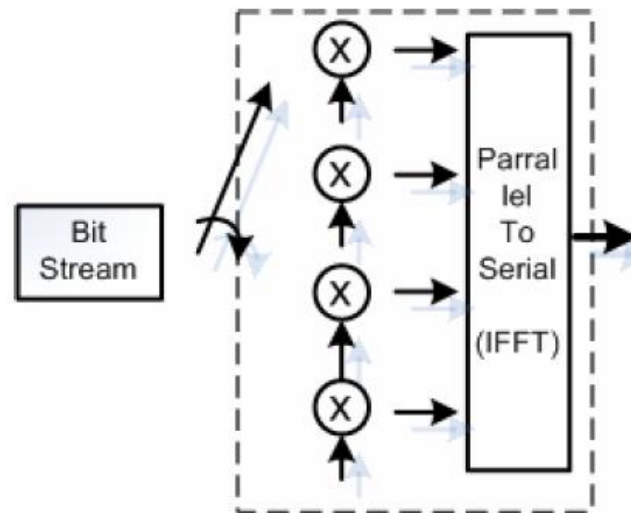


# OFDM

- It is possible to use any other modulation method, QPSK, 8PSK 32-QAM...
  - There is no limit on what modulation to use
  - We can even use TCM which provides coding in addition to modulation
- Carrier 1 - We need to transmit 1, 1, 1 -1, -1, -1 on the BPSK carrier of frequency 1 Hz
- Carrier 2 - The next carrier is of frequency 2 Hz. It is the next orthogonal/harmonic to frequency of the first carrier of 1 Hz. The bits in the second column (c2) are: 1, 1, -1, 1, 1, -1

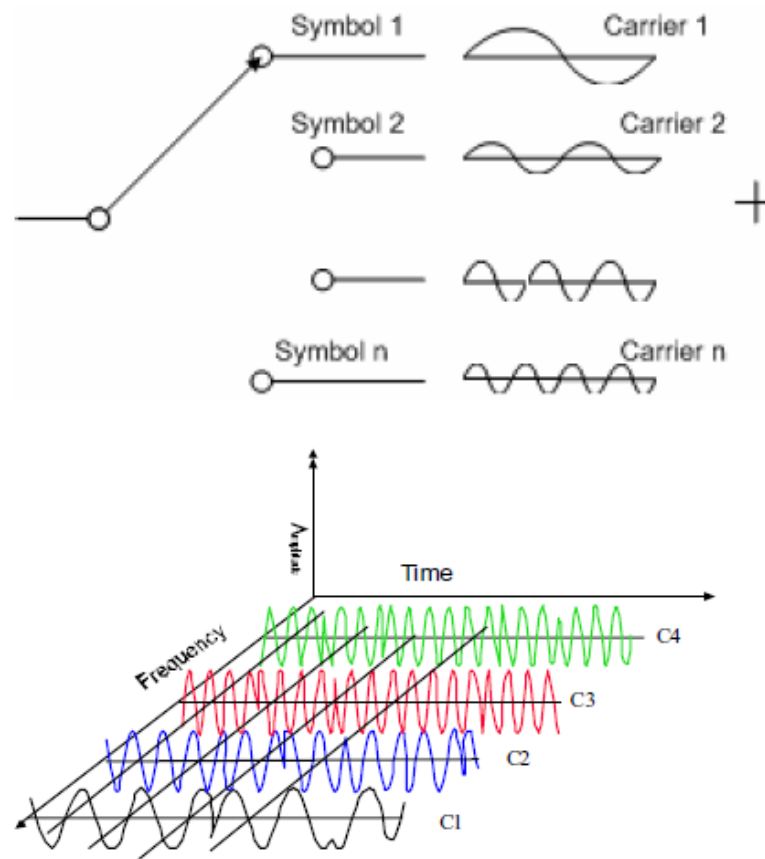
# OFDM

- Carrier 3/4 - Carrier 3 frequency is equal to 3 Hz and Carrier 4 has a frequency of 4 Hz. The 3rd carrier is modulated with: -1, 1, 1, -1, -1, 1 and the 4th with: -1, -1, -1, -1, -1, -1, 1
- Finally, we have to add all four of these modulated carriers to create the OFDM signal, often produced by a block called the IFFT

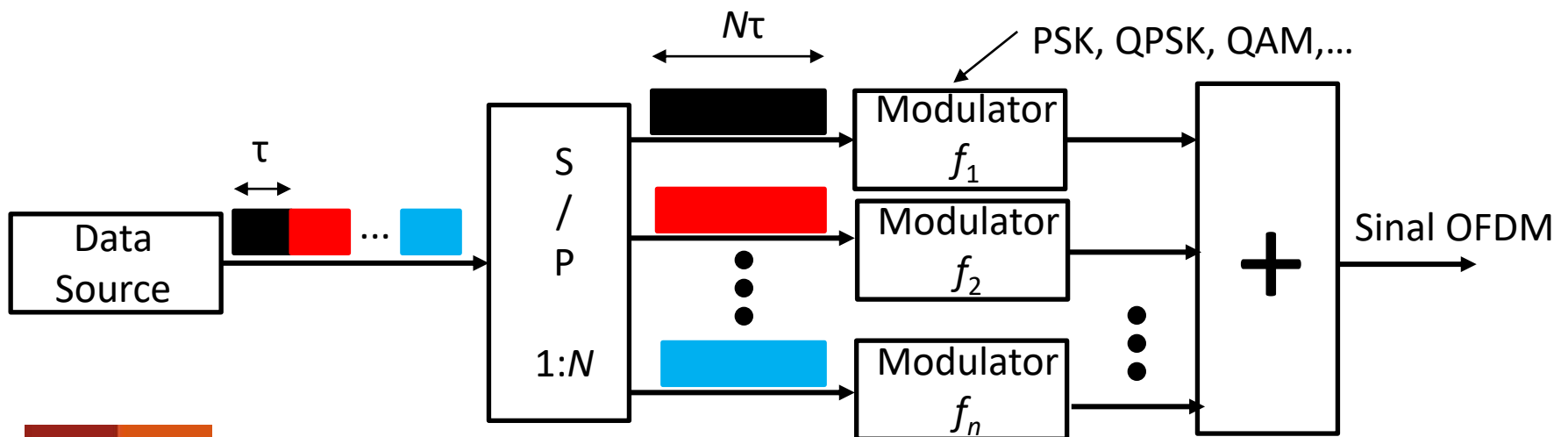
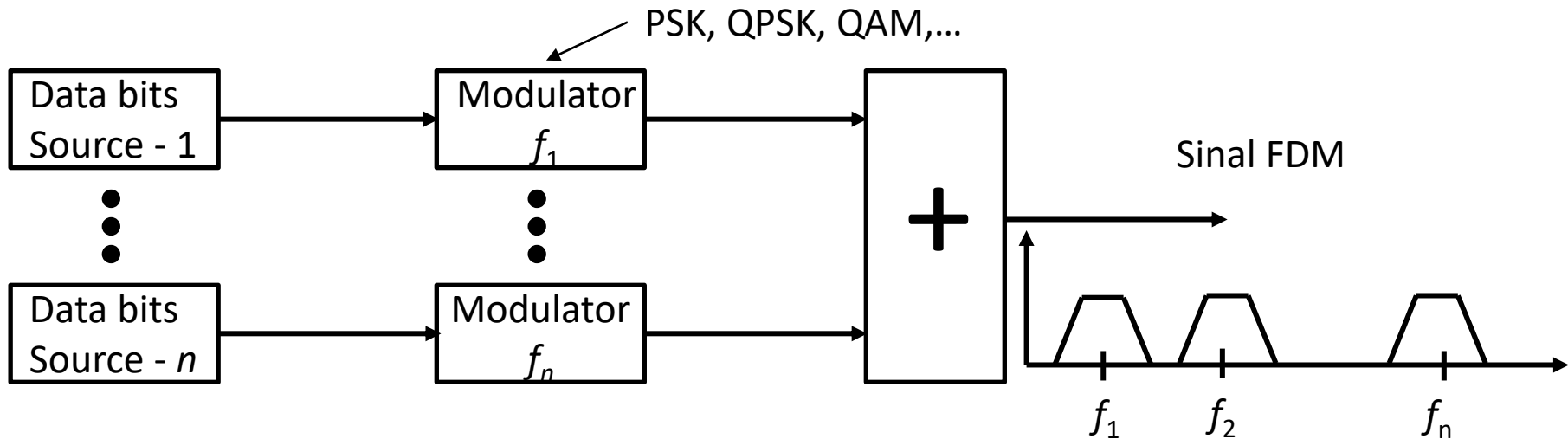


# OFDM

- OFDM signal in time and frequency domain



# FDM / OFDM



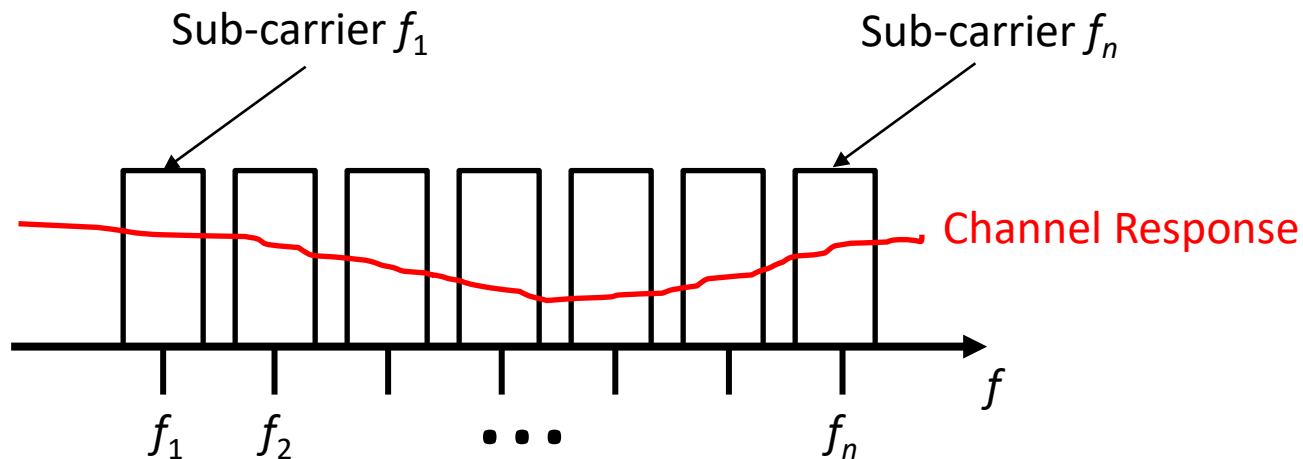
# OFDM

- If we have to transmit a high data rate sequence:
  - First, we have to divide into several sequences with lower rate
  - Each of these sequences modulates a sub-carrier in contrast to the single carrier modulation used in FDM
  - The duration of the impulse of each subcarrier is  $K$  times larger the original bit
  - The bandwidth of every modulated sub-carrier is equal to  $1/K$  of single carrier



# OFDM

- Channel Frequency Response



- Vary significantly, but little in every sub-band
  - Signal distortion is moderate
  - Equalization (Distortion compensation) - Unnecessary

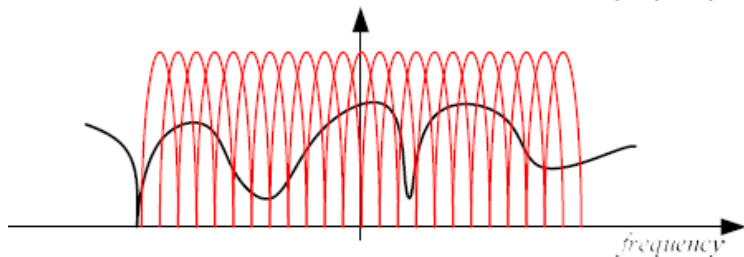
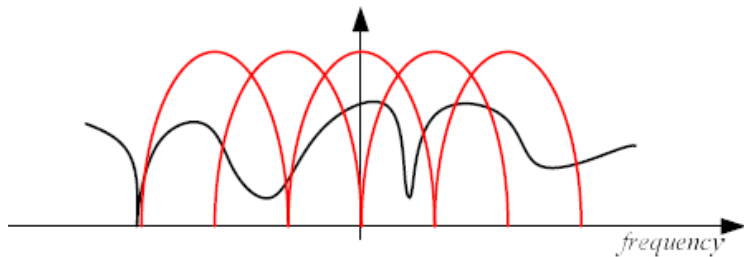
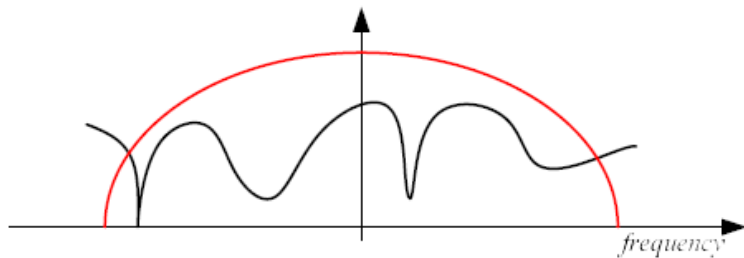
# OFDM

- Channel Frequency Response
  - For the same bit rate the symbol period increases as the number of carriers increases
  - The symbol period is longer than the channel impulse response
  - The channel impulse response only affects the beginning of each symbol
  - Parallel transmission overcomes the multipath problem



# OFDM

- Channel Frequency Response



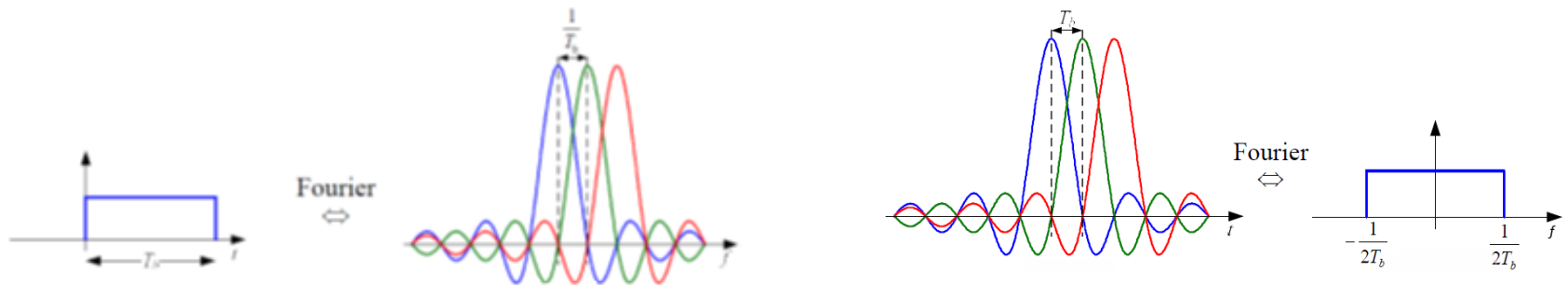
- As the number of carriers increase their bandwidth becomes narrow

- Each carrier only “sees” a small fraction of the total occupied spectrum

- For each carrier the channel transfer function can be considered “flat”

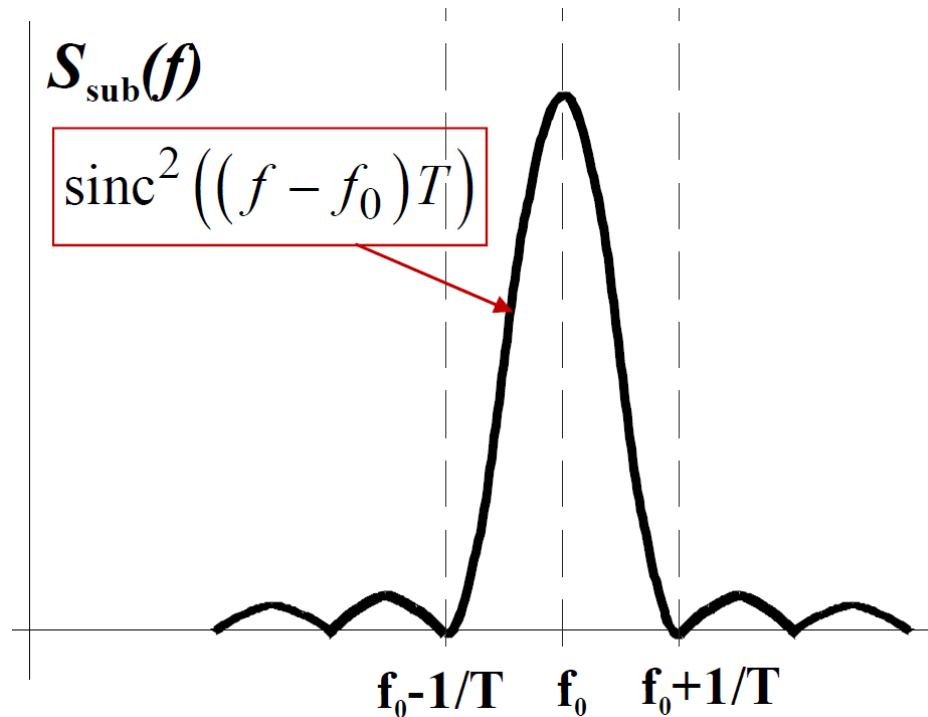
# OFDM

- Channel Frequency Response
  - A Nyquist pulse has the smallest bandwidth for a given bit rate
  - In OFDM we have Nyquist pulses with the time and frequency domain interchanged



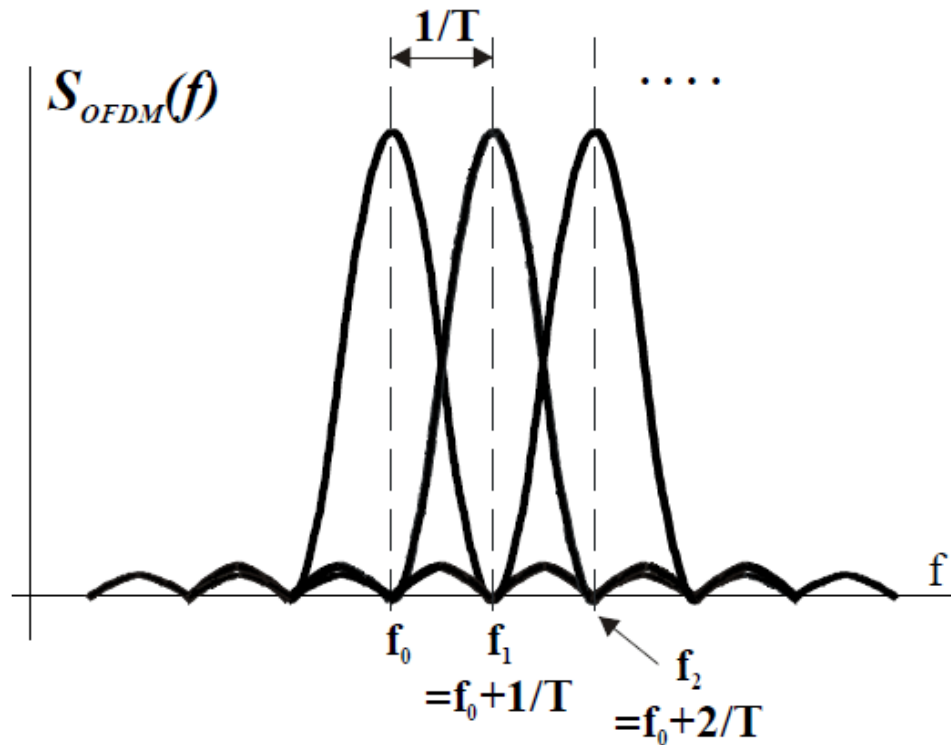
# OFDM

- Channel Frequency Response
  - Power Spectral Density (PSD) of a Sub-carrier



# OFDM

- Channel Frequency Response
  - PSD of OFDM signal



# OFDM

- Channel Frequency Response
  - In OFDM there are an overlap of PSDs of the various sub-carriers
  - There are no guard bands
  - Spectral efficiency increases
- If PSDs overlay how to separate information from the various sub-carriers?

# OFDM

- Channel Frequency Response
  - The answer is in the  $1/T$  separation between subcarriers
  - $T$  - duration of the band base pulse in each sub-carrier
  - $r = 1/T$  symbols/sec

$$f_n = f_0 + n\Delta f, \quad n = 0, 1, \dots, K-1 \quad \Delta f = \frac{1}{T} = r$$

$$\int_0^T \cos(2\pi f_j t) \times \cos(2\pi f_l t) dt = 0, \quad \text{for all } j, l, j \neq l$$

➤ Sub-carriers are orthogonal



# OFDM

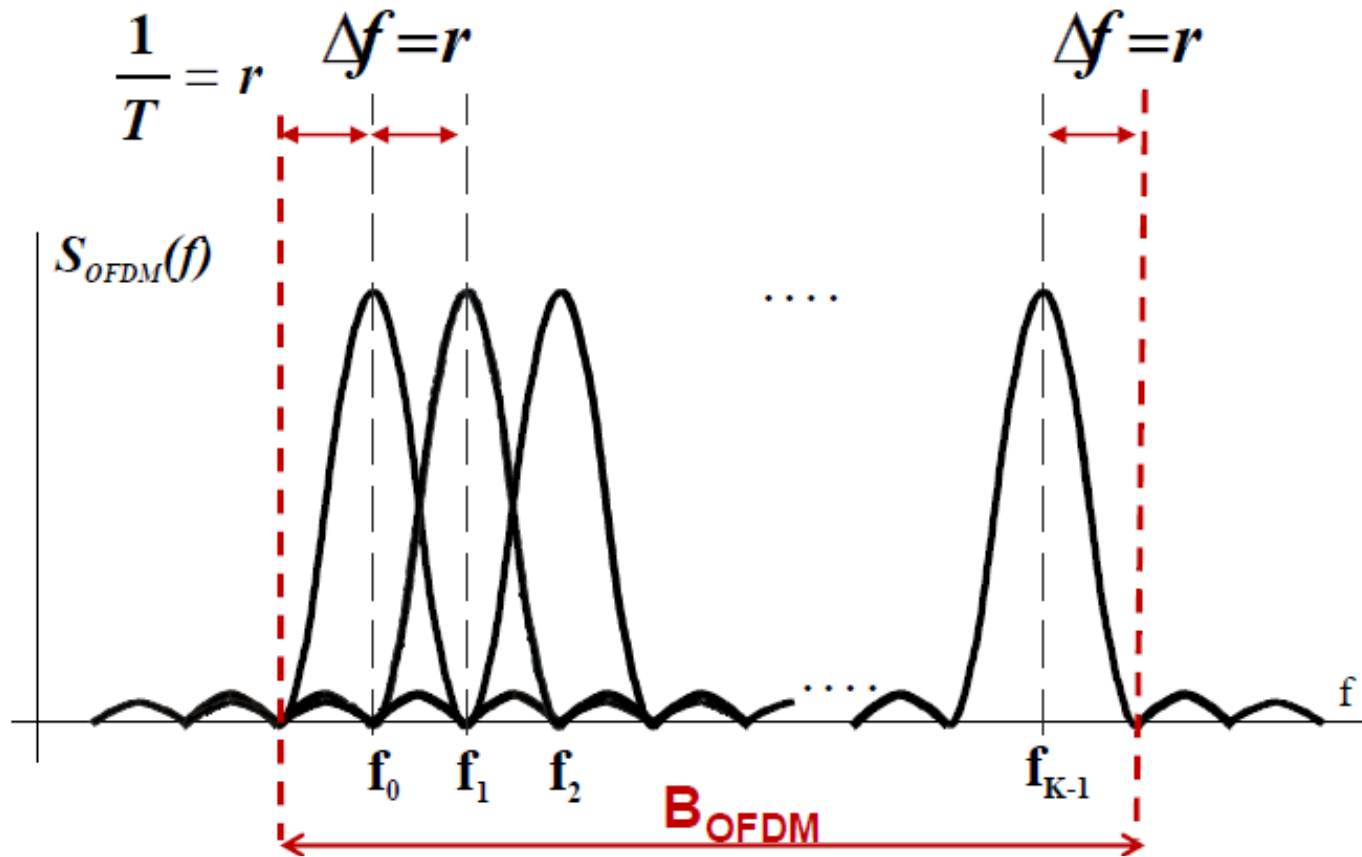
- Channel Frequency Response
  - If we have different information on 2 sub-carriers:

$$\begin{aligned} & \int_0^T \left[ a \cos(2\pi f_j t) + b \cos(2\pi f_l t) \right] \times \cos(2\pi f_j t) dt \\ &= \int_0^T a \cos^2(2\pi f_j t) dt + b \int_0^T \cancel{\cos(2\pi f_j t) \cos(2\pi f_l t)} dt \\ &= \frac{a}{2} \int_0^T dt + \frac{a}{2} \int_0^T \cos(4\pi f_j t) dt = \frac{aT}{2} \end{aligned}$$

- In the receiver it will be possible to separate the information transported in the subcarrier with frequency  $f_j$

# OFDM

- OFDM Bandwidth





# OFDM

- OFDM Bandwidth

- Considering the first null of the  $\text{sinc}^2()$  function to define the bandwidth of a signal, the bandwidth of the OFDM signal is given by:

$$B_{OFDM} = (K - 1)r + 2r = (K + 1)r$$

$$B_{OFDM} = (K + 1)r = \frac{(K+1)r_b}{nK} = \frac{(K+1)r_b}{K \log_2 M}$$

- $r$  = symbol rate of each sub-carrier
- $K$  = number of sub-carriers
- $M$  = number of constellation points for each sub-carrier



# OFDM

- OFDM Bandwidth versus single carrier bandwidth
  - In this case:  $K = 1$
  - $B_{SC} = 2r_b/n$
  - $B_{OFDM} / B_{SC} = (K + 1) / 2K \approx 1/2, K \gg 1$
- Spectral efficiency:

$$\eta_{OFDM} = \frac{r_b}{B_{OFDM}} = \frac{Kn}{(K + 1)} \approx n \text{ bits / seg / Hz, } K \gg 1$$



# OFDM

- OFDM Bandwidth

- Bandwidth of a single carrier

- $B_{SC} = 2r_b/n$

- Spectral efficiency of single carrier:

$$\eta_{port} = \frac{r_b}{B_{SC}} = \frac{n}{2} \text{ bits / seg / Hz}$$

- Spectral efficiency OFDM versus single carrier:

$$\frac{\eta_{OFDM}}{\eta_{SC}} = \frac{\frac{Kn}{(K+1)}}{\frac{n}{2}} = \frac{2K}{(K+1)} \approx 2, \quad K \gg 1$$



# OFDM

- Example:
  - Example: We want to transmit at a rate of 1600 bit/sec in an OFDM system that uses 8 sub-carriers, each of which is modulated in QPSK. Knowing that the lowest frequency of the sub-carriers is 650 Hz, determine:
    - (a) The highest frequency of the sub-carriers
    - (b) The bandwidth of the OFDM signal
    - (c) The bandwidth required if a single carrier was used



# OFDM

- Example (Solution):

- (a) Each sub-carrier:

$$\frac{1600}{8} = 200 \text{ bit/s} \quad r = \frac{200}{2} = 100 \text{ symbols/s (QPSK)}$$

$$\Delta f = r = 100 \text{ Hz}$$

$$f_k = f_0 + k\Delta f$$



$$f_7 = 650 + 7 \times 100 = 1350 \text{ Hz}$$

- (b) The bandwidth of the OFDM signal

$$B_{OFDM} = (K + 1)r = (8 + 1) \times 100 = 900 \text{ Hz}$$

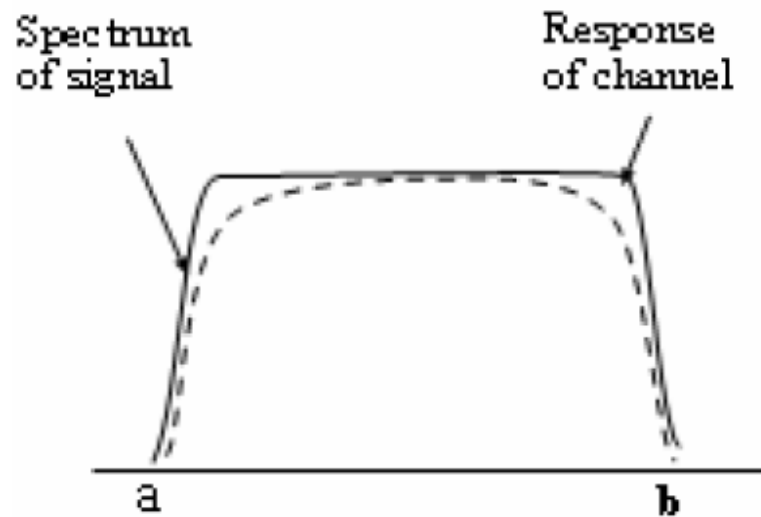
- (c) The bandwidth required if a single carrier was used

$$B_{sc} = 2r_b / n = 2 \times 1600 / 2 = 1600 \text{ Hz}$$



# OFDM

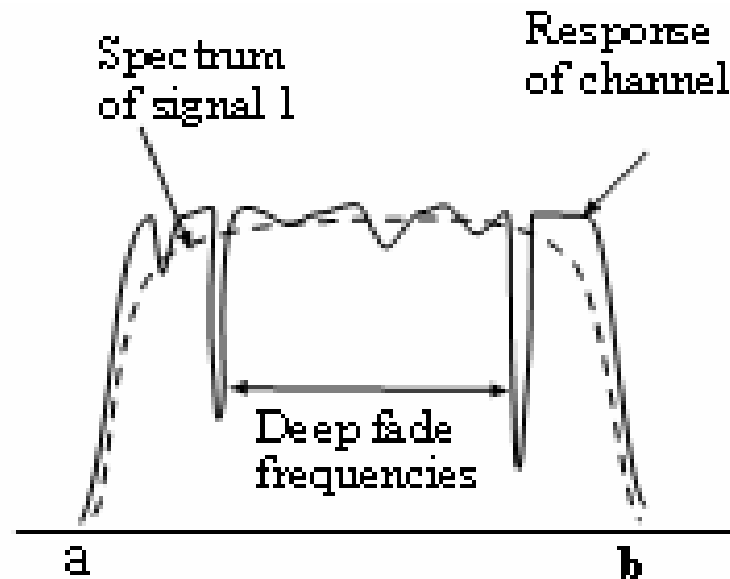
- Channel Frequency Response



- The signal we want to send and the channel frequency response are well matched

# OFDM

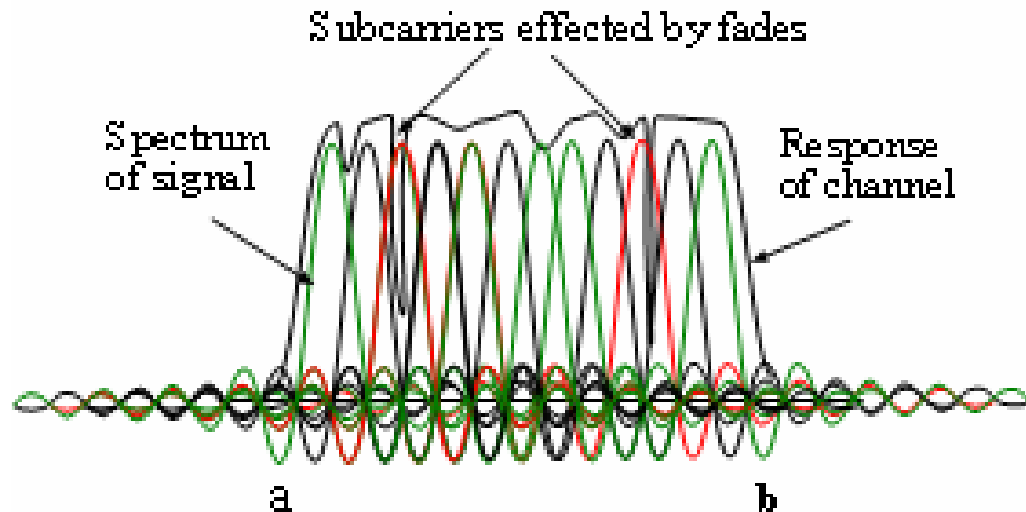
- Channel Frequency Response



- A fading channel has frequencies that do not allow anything to pass. Data is lost sporadically

# OFDM

- Channel Frequency Response



- With OFDM, where we have many little sub-carriers, only a small sub-set of the data is lost due to fading



# OFDM

- Channel Frequency Response
  - In this example, an OFDM signal cross a frequency-selective response channel and only two sub-carriers are affected, all the others are perfectly OK
  - Instead of the whole symbol being knocked out, we lose just a small subset of the  $(1/N)$  bits
  - With proper channel coding, this can be recovered



# Reading Material

- Henrik Schulze and Christian Luders, Theory and Applications of OFDM and CDMA, Wideband Wireless Communications, John Wiley & Sons Inc., 2005 (ISBN-13 978-0-470-85069-5)
- Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)



# Reading Material

